

# Aortic Stenosis with A Fib and CHF



## Section I: Scenario Demographics

|                        |                                                               |                                                       |                                     |
|------------------------|---------------------------------------------------------------|-------------------------------------------------------|-------------------------------------|
| Scenario Title:        | Atrial Fibrillation and CHF in a patient with Aortic Stenosis |                                                       |                                     |
| Date of Development:   | 20/04/2016 (DD/MM/YYYY)                                       |                                                       |                                     |
| Target Learning Group: | <input type="checkbox"/> Juniors (PGY 1 – 2)                  | <input checked="" type="checkbox"/> Seniors (PGY ≥ 3) | <input type="checkbox"/> All Groups |

## Section II: Scenario Developers

|                              |                            |
|------------------------------|----------------------------|
| Scenario Developer(s):       | Donika Orlich              |
| Affiliations/Institution(s): | McMaster University        |
| Contact E-mail (optional):   | Donika.orlich@medportal.ca |

## Section III: Curriculum Integration

| Learning Goals & Objectives |                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Educational Goal:           | Expose learners to a patient with decompensated AS due to A. fib resulting in CHF                                                                                                                                                                                                                                                                                                                                     |
| CRM Objectives:             | 1. Demonstrates situational awareness by anticipating likely events and communicating these clearly with all team members                                                                                                                                                                                                                                                                                             |
| Medical Objectives:         | 1. Demonstrates strong communication skills by communicating risks and benefits of procedural sedation and cardioversion with the patient and conducting a code discussion<br>2. Initiates broad work-up for acute decompensation in the patient with AS<br>3. Employs a tailored approach to hemodynamic resuscitation in the AS patient<br>4. Employs a tailored approach to the management of CHF in an AS patient |

### Case Summary: Brief Summary of Case Progression and Major Events

A 78-year-old male presents with increased SOB over the past 4 days. A recent ECHO will be presented showing severe AS. The ECG will demonstrate new A Fib with a HR of 150 and the CXR will show CHF. The patient will be normotensive, at first, but will become hypotensive shortly after. The team will then need to decide whether to cardiovert the patient or attempt rate control. If these are done safely, the patient will respond and then develop worsening CHF. Definitive management should be sought with early cardiology/cardiac surgery consult. If management is not carried out judiciously, the patient will become profoundly hypotensive.

### References

Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby.

ACC/AHA/ESC 2006 guidelines for the management of patients with atrial fibrillation: Article accessed April 20, 2016.  
<http://europace.oxfordjournals.org/content/8/9/651>



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## Section IV: Scenario Script

| A. Scenario Cast & Realism                              |                                                                                                                                                               |                                                         |                                                 |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|-------------------------------------------------|
| Patient:                                                | <input checked="" type="checkbox"/> Computerized Mannequin                                                                                                    | Realism:                                                | <input checked="" type="checkbox"/> Conceptual  |
|                                                         | <input type="checkbox"/> Mannequin                                                                                                                            |                                                         | <input type="checkbox"/> Physical               |
|                                                         | <input type="checkbox"/> Standardized Patient                                                                                                                 |                                                         | <input type="checkbox"/> Emotional/Experiential |
|                                                         | <input type="checkbox"/> Hybrid                                                                                                                               |                                                         | <input type="checkbox"/> Other:                 |
|                                                         | <input type="checkbox"/> Task Trainer                                                                                                                         |                                                         | <input type="checkbox"/> N/A                    |
| <i>Select most important dimension(s)</i>               |                                                                                                                                                               |                                                         |                                                 |
| Confederates                                            |                                                                                                                                                               |                                                         |                                                 |
| Brief Description of Role                               |                                                                                                                                                               |                                                         |                                                 |
| Wife                                                    | Provides latest ECHO report from cardiologist. Present for code discussion. If no consent obtained prior to procedural sedation, she will ask about the risk. |                                                         |                                                 |
|                                                         |                                                                                                                                                               |                                                         |                                                 |
| B. Required Monitors                                    |                                                                                                                                                               |                                                         |                                                 |
| <input checked="" type="checkbox"/> EKG Leads/Wires     | <input type="checkbox"/> Temperature Probe                                                                                                                    | <input type="checkbox"/> Central Venous Line            |                                                 |
| <input checked="" type="checkbox"/> NIBP Cuff           | <input checked="" type="checkbox"/> Defibrillator Pads                                                                                                        | <input type="checkbox"/> Capnography                    |                                                 |
| <input checked="" type="checkbox"/> Pulse Oximeter      | <input type="checkbox"/> Arterial Line                                                                                                                        | <input type="checkbox"/> Other:                         |                                                 |
| C. Required Equipment                                   |                                                                                                                                                               |                                                         |                                                 |
| <input checked="" type="checkbox"/> Gloves              | <input checked="" type="checkbox"/> Nasal Prongs                                                                                                              | <input type="checkbox"/> Scalpel                        |                                                 |
| <input checked="" type="checkbox"/> Stethoscope         | <input checked="" type="checkbox"/> Venturi Mask                                                                                                              | <input type="checkbox"/> Tube Thoracostomy Kit          |                                                 |
| <input checked="" type="checkbox"/> Defibrillator       | <input checked="" type="checkbox"/> Non-Rebreather Mask                                                                                                       | <input type="checkbox"/> Cricothyroidotomy Kit          |                                                 |
| <input checked="" type="checkbox"/> IV Bags/Lines       | <input checked="" type="checkbox"/> Bag Valve Mask                                                                                                            | <input type="checkbox"/> Thoracotomy Kit                |                                                 |
| <input checked="" type="checkbox"/> IV Push Medications | <input checked="" type="checkbox"/> Laryngoscope                                                                                                              | <input type="checkbox"/> Central Line Kit               |                                                 |
| <input type="checkbox"/> PO Tabs                        | <input type="checkbox"/> Video Assisted Laryngoscope                                                                                                          | <input type="checkbox"/> Arterial Line Kit              |                                                 |
| <input type="checkbox"/> Blood Products                 | <input checked="" type="checkbox"/> ET Tubes                                                                                                                  | <input checked="" type="checkbox"/> Other: <b>BiPap</b> |                                                 |
| <input type="checkbox"/> Intraosseous Set-up            | <input type="checkbox"/> LMA                                                                                                                                  | <input type="checkbox"/> Other:                         |                                                 |
| D. Moulage                                              |                                                                                                                                                               |                                                         |                                                 |
| Old man mask.                                           |                                                                                                                                                               |                                                         |                                                 |
| E. Approximate Timing                                   |                                                                                                                                                               |                                                         |                                                 |
| Set-Up:                                                 | 5 min                                                                                                                                                         | Scenario:                                               | 15 min                                          |
|                                                         |                                                                                                                                                               | Debriefing:                                             | 20 min                                          |

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## Section V: Patient Data and Baseline State

| A. Clinical Vignette: To Read Aloud at Beginning of Case                                                                                                                                                                                                                  |                                                                                           |                                                                                                                                                                |                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| A 78-year-old male presents via EMS with 4 days of increased SOB. The triage nurse comes to tell you she has put him in the resuscitation bay due to unstable vitals. HR was in the 150s. The O <sub>2</sub> SAT was 86% on RA when EMS arrived, but is now 95% on a NRB. |                                                                                           |                                                                                                                                                                |                             |
| B. Patient Profile and History                                                                                                                                                                                                                                            |                                                                                           |                                                                                                                                                                |                             |
| Patient Name: Geoff Miller                                                                                                                                                                                                                                                |                                                                                           | Age: 78                                                                                                                                                        | Weight: 80kg                |
| Gender: <input checked="" type="checkbox"/> M <input type="checkbox"/> F                                                                                                                                                                                                  |                                                                                           | Code Status: FULL                                                                                                                                              |                             |
| Chief Complaint: SOB                                                                                                                                                                                                                                                      |                                                                                           |                                                                                                                                                                |                             |
| History of Presenting Illness: Increased SOB over the past 4 days. This morning he was severely SOB on awaking therefore wife called EMS.                                                                                                                                 |                                                                                           |                                                                                                                                                                |                             |
| Past Medical History:                                                                                                                                                                                                                                                     | "Valve Problem"                                                                           | Medications:                                                                                                                                                   | Enalapril 5mg PO daily      |
|                                                                                                                                                                                                                                                                           | HTN                                                                                       |                                                                                                                                                                | Simvastatin 20mg PO daily   |
|                                                                                                                                                                                                                                                                           | Dyslipidemia                                                                              |                                                                                                                                                                | Metformin 500mg PO BID      |
|                                                                                                                                                                                                                                                                           | DM 2                                                                                      |                                                                                                                                                                |                             |
| Allergies: None                                                                                                                                                                                                                                                           |                                                                                           |                                                                                                                                                                |                             |
| Social History: Lives with wife. No smoking. No EtOH                                                                                                                                                                                                                      |                                                                                           |                                                                                                                                                                |                             |
| Family History: Nil.                                                                                                                                                                                                                                                      |                                                                                           |                                                                                                                                                                |                             |
| Review of Systems:                                                                                                                                                                                                                                                        | CNS:                                                                                      | No syncope/presyncope                                                                                                                                          |                             |
|                                                                                                                                                                                                                                                                           | HEENT:                                                                                    | Nil                                                                                                                                                            |                             |
|                                                                                                                                                                                                                                                                           | CVS:                                                                                      | Mild chest pressure last 4 days off and on. Worse on exertion. No radiation. No pedal edema. No palpitations. No history of angina, CHF or exertional syncope. |                             |
|                                                                                                                                                                                                                                                                           | RESP:                                                                                     | Worsening SOB over past 4d. + orthopnea. No cough/sputum                                                                                                       |                             |
|                                                                                                                                                                                                                                                                           | GI:                                                                                       | Nil                                                                                                                                                            |                             |
|                                                                                                                                                                                                                                                                           | GU:                                                                                       | Nil                                                                                                                                                            |                             |
|                                                                                                                                                                                                                                                                           | MSK:                                                                                      | No calf swelling/pain.                                                                                                                                         | INT:                        |
| C. Baseline Simulator State and Physical Exam                                                                                                                                                                                                                             |                                                                                           |                                                                                                                                                                |                             |
| <input type="checkbox"/> No Monitor Display                                                                                                                                                                                                                               |                                                                                           | <input checked="" type="checkbox"/> Monitor On, no data displayed                                                                                              |                             |
| <input type="checkbox"/> Monitor on Standard Display                                                                                                                                                                                                                      |                                                                                           |                                                                                                                                                                |                             |
| HR: 150/min                                                                                                                                                                                                                                                               | BP: 105/50                                                                                | RR: 30/min                                                                                                                                                     | O <sub>2</sub> SAT: 95% NRB |
| Rhythm: <b>A.fib</b>                                                                                                                                                                                                                                                      | T: 36.6°C                                                                                 | Glucose: 10 mmol/L                                                                                                                                             | GCS: 15 (E4 V5 M6)          |
| General Status: Increased WOB but no significant distress.                                                                                                                                                                                                                |                                                                                           |                                                                                                                                                                |                             |
| CNS:                                                                                                                                                                                                                                                                      | Nil                                                                                       |                                                                                                                                                                |                             |
| HEENT:                                                                                                                                                                                                                                                                    | Nil                                                                                       |                                                                                                                                                                |                             |
| CVS:                                                                                                                                                                                                                                                                      | <b>Grade 3 SEM (crescendo-decrescendo) with rad to carotids. Pulsus parvus et tardus.</b> |                                                                                                                                                                |                             |
| RESP:                                                                                                                                                                                                                                                                     | <b>Bilateral crackles</b>                                                                 |                                                                                                                                                                |                             |
| ABDO:                                                                                                                                                                                                                                                                     | Soft. No ascites.                                                                         |                                                                                                                                                                |                             |
| GU:                                                                                                                                                                                                                                                                       | Nil                                                                                       |                                                                                                                                                                |                             |
| MSK:                                                                                                                                                                                                                                                                      | No pedal edema                                                                            | SKIN:                                                                                                                                                          | Normal                      |

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## Section VI: Scenario Progression

### Scenario States, Modifiers and Triggers

| Patient State                                                                                                                     | Patient Status                                        | Learner Actions, Modifiers & Triggers to Move to Next State                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Baseline State</b><br>Rhythm: A. fib<br>HR: 150/min<br>BP: 105/50<br>RR: 30/min<br>O <sub>2</sub> SAT: 95% NRB<br>T: 36.6°C | Breathing quickly.<br>Speaking in 3-4 word sentences. | <u>Learner Actions</u><br><input type="checkbox"/> IV, O <sub>2</sub> , monitors<br><input type="checkbox"/> Labs (troponin, TSH, extended lytes, VBG, basic labs)<br><input type="checkbox"/> Portable CXR<br><input type="checkbox"/> ECG<br><input type="checkbox"/> ASA 160mg PO<br><input type="checkbox"/> Consider early BiPAP<br><input type="checkbox"/> Code discussion | <u>Modifiers</u><br><i>Changes to patient condition based on learner action</i><br>- Ask about PMHx → wife gives copy of last ECHO<br><br><u>Triggers</u><br><i>For progression to next state</i><br>- BiPAP started, preparing meds for rate control, or 5 min into case → <b>2. Hypotension</b><br>- Decision to cardiovert → <b>3. Cardioversion</b>                                                                                                                                                  |
| <b>2. Hypotension</b><br>Rhythm: A. fib<br>HR: 150/min<br>BP: 80/60<br>RR: 20/min<br>O <sub>2</sub> SAT: 98% on 50%               | Decreased work of breathing on BiPAP.                 | <u>Learner Actions</u><br><input type="checkbox"/> 250cc NS bolus<br><input type="checkbox"/> Repeat 250cc NS bolus<br><input type="checkbox"/> Prepare for cardioversion<br><input type="checkbox"/> ± Low-dose norepi (0.03 mg/kg/min)<br><input type="checkbox"/> ± Amiodarone or digoxin for rate control<br><input type="checkbox"/> Consult cardiology +/- cardiac surgery  | <u>Modifiers</u><br>- 250cc bolus → BP 85/65, HR 145, O <sub>2</sub> 96%<br>- 500cc total bolus → BP 90/70, HR 140, O <sub>2</sub> 90%<br>- Norepi → BP 100/75, HR 140<br>- Amiodarone/dig given → HR 100, BP 100/75<br>- BB/CCB used for rate control → HR 100, BP 70/60<br>- If no rate control/cardioversion by 5 min → RN to prompt "what's the plan for his HR?"<br><br><u>Triggers</u><br>- Decision to cardiovert → <b>3. Cardioversion</b><br>- Rate control attempted → <b>4. Worsening CHF</b> |
| <b>3. Cardioversion</b><br>Vitals as per previous state                                                                           | Patient concerned but agrees to cardioversion.        | <u>Learner Actions</u><br><input type="checkbox"/> Consent for cardioversion<br><input type="checkbox"/> Consider small (250cc) fluid bolus prior to sedation<br><input type="checkbox"/> Push-dose phenylephrine at bedside<br><input type="checkbox"/> ½ dose sedation (fentanyl ± midazolam)<br><input type="checkbox"/> Synchronized shock                                    | <u>Modifiers</u><br>- Shock at <200J → no change<br>- Shock at 200J → converts to sinus rhythm HR 82, BP 100/80<br><br><u>Triggers</u><br>- Sinus rhythm achieved → <b>4. Worsening CHF</b>                                                                                                                                                                                                                                                                                                              |



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|                                                                                                                                                                                                  |                                                             |                                                                                                                                                                                                                                                                                                    |                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>4. Worsening CHF</b><br>Rhythm: a fib or NSR<br>(as per previous state)<br>HR: 90<br>BP: 100/80<br>O <sub>2</sub> SAT: 90% on 100%<br>RR: 30<br><br><b>**Labs given at beginning of state</b> | Increased WOB.<br><br>Diaphoretic.<br><br>Audible crackles. | <u>Learner Actions</u><br>- <input type="checkbox"/> BiPAP (if not already)<br>- <input type="checkbox"/> Lasix (10mg-40mg IV)<br>- <input type="checkbox"/> Cardiac surgery consult<br>- <input type="checkbox"/> Cardiology consult<br>- <input type="checkbox"/> Anticipate need for intubation | <u>Modifiers</u><br>- Nitro given → BP 70/40<br><br><u>Triggers</u><br>END CASE PRN |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|

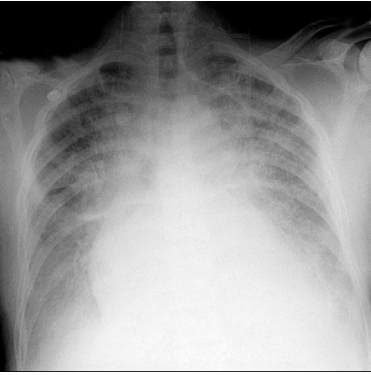
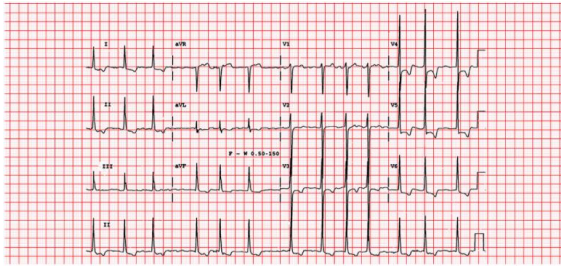
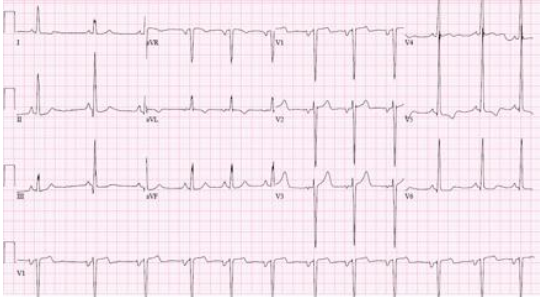


# Aortic Stenosis with A Fib and CHF



## Section VII: Supporting Documents, Laboratory Results, & Multimedia

| Laboratory Results |          |                       |                       |                        |              |             |
|--------------------|----------|-----------------------|-----------------------|------------------------|--------------|-------------|
| Na: 140            | K: 4     | Cl: 105               | HCO <sub>3</sub> : 25 | BUN: 5                 | Cr: 80       | Glu: 5.5    |
| Ca: 2.50           |          | Mg: 0.80              |                       | PO <sub>4</sub> : 1.20 |              | Albumin: 35 |
|                    |          |                       |                       |                        |              |             |
| VBG                | pH: 7.30 | PCO <sub>2</sub> : 35 | PO <sub>2</sub> : 95  | HCO <sub>3</sub> : 25  | Lactate: 2.0 |             |
| Troponin I: 40     |          |                       | TSH: 2.5              |                        |              |             |
| WBC: 5.0           |          | Hg: 120               | Hct: 0.4              |                        | Plt: 300     |             |
|                    |          |                       |                       |                        |              |             |

| Images (ECGs, CXRs, etc.)                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>CXR: CHF</p>  <p>CXR source: <a href="https://www.med-ed.virginia.edu/courses/rad/cxr/pathology2Bchest.html">https://www.med-ed.virginia.edu/courses/rad/cxr/pathology2Bchest.html</a></p> | <p>ECG: LVH + A.fib</p>  <p>ECG source: <a href="http://www.wikidoc.org/index.php/Atrial_fibrillation_EKG_examples">http://www.wikidoc.org/index.php/Atrial_fibrillation_EKG_examples</a></p>                                   |
| <p>Lung U/S: B lines bilaterally (CHF)</p>                                                                                                                                                                                                                                     | <p>ECG: NSR with LVH</p>  <p>ECG source: <a href="http://bestpractice.bmj.com/best-practice/monograph/409/resources/image/bp/5.html">http://bestpractice.bmj.com/best-practice/monograph/409/resources/image/bp/5.html</a></p> |

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## Section VIII: Debriefing Guide

| General Debriefing Plan                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> Individual                                                                                                                                                                                                                                                                                                                                  | <input checked="" type="checkbox"/> Group                                                                                                                                                                                                                                                                                                                                                                             |
| <input type="checkbox"/> With Video                                                                                                                                                                                                                                                                                                                                  | <input checked="" type="checkbox"/> Without Video                                                                                                                                                                                                                                                                                                                                                                     |
| Objectives                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Educational Goal:                                                                                                                                                                                                                                                                                                                                                    | Expose learners to a patient with decompensated AS due to A. fib resulting in CHF                                                                                                                                                                                                                                                                                                                                     |
| CRM Objectives:                                                                                                                                                                                                                                                                                                                                                      | 1. Demonstrates situational awareness by anticipating likely events and communicating these clearly with all team members                                                                                                                                                                                                                                                                                             |
| Medical Objectives:                                                                                                                                                                                                                                                                                                                                                  | 1. Demonstrates strong communication skills by communicating risks and benefits of procedural sedation and cardioversion with the patient and conducting a code discussion<br>2. Initiates broad work-up for acute decompensation in the patient with AS<br>3. Employs a tailored approach to hemodynamic resuscitation in the AS patient<br>4. Employs a tailored approach to the management of CHF in an AS patient |
| Sample Questions for Debriefing                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 1) What was your approach to managing the A.fib in this patient? Were they stable or unstable?<br>2) What factors did you consider when deciding whether or not to cardiovert this patient?<br>3) What modifications to your management of CHF did you make for this patient with AS? Why?<br>4) What is your approach to managing hypotension in a patient with AS? |                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Key Moments                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 1) Recognition of unstable a fib and need for cardioversion.                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 2) Tailored approach to hypotension and CHF in the patient with AS                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 3) Safe procedural sedation planning                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                       |

# Aortic Stenosis with A Fib and CHF



## Supplemental ECHO

### Cardiac Ultrasound Unit

Patient Name: Geoff Miller

Age: 78

Sex: Male

Location: outpatient

Indication: Reassess aortic stenosis

### M-MODE AND TWO DIMENSIONAL FINDINGS

Left ventricular size is increased. Left ventricular function is slightly reduced, with an estimated EF of 35%. Septal motion is normal. The left atrium is mildly dilated. Right ventricular function is normal. There is no pericardial effusion.

### SPECTRAL AND COLOR FLOW DOPPLER FINDINGS

There is moderate aortic stenosis with aortic root diameter measured at 0.9 cm. There is decreased maximum Aortic Velocity measured at 3mmHg, with a mean pressure gradient of 30mmHg. There is no mitral or tricuspid regurgitation.

### CONCLUSION

Reduced left ventricular function with an EF of 35%

Dilated left ventricle

Dilated left atrium

Severe aortic stenosis with root diameter of 0.9 cm

No pulmonary hypertension

Compared to study 2 years ago, there is worsening of aortic stenosis with increased mean peak pressure gradient and decreased aortic root diameter